



ICPAC

Data systems for Early warning

Interoperable Hydrometeorological Data Systems for Multi-Hazard Early
Warning Systems
14-17 Addis Ababa

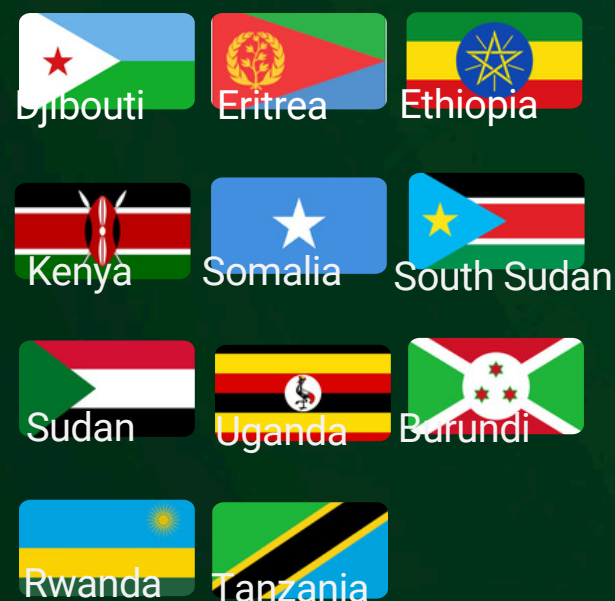
► WHAT IS ICPAC?

ICPAC is a Regional Climate Centre of the Greater Horn of Africa, with the mandate to provide regional climate services and early warning support to member states

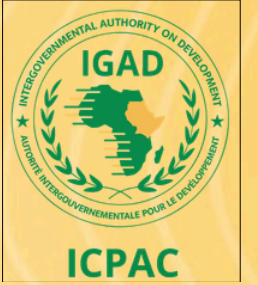
ICPAC is accredited by the WMO as a **.Regional Climate Center (RCC)**

One of the key mandatory functions of ICPAC-RCC is establish an operational data services, to support operational seasonal prediction and climate monitoring

ICPAC Member states



► ICPAC SERVICES



Climate
Forecasting



Disaster Risk
Management



Water
Resources



Environmental
Monitoring



Capacity
Development



Climate
Change



Climate Peace
and Security



Climate Information
Dissemination



Agriculture and
Food Security

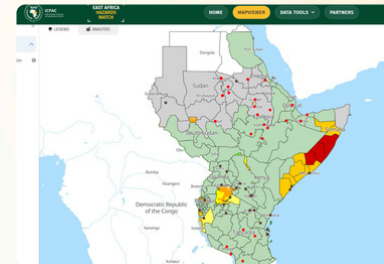
Data and information platforms & Digital Innovations



01

East Africa Hazards Watch

<https://eahazardswatch.icpac.net/>

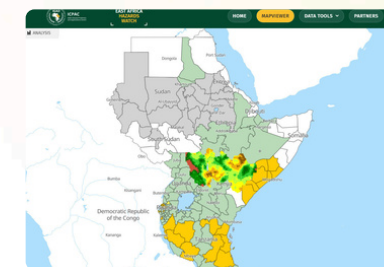


A near-real-time regional monitoring system providing early warning on climate and weather hazards across Eastern Africa.

02

East Africa Agriculture Watch

<https://agriculturehotspots.icpac.net/>

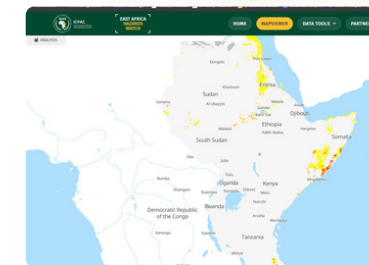


A digital monitoring tool providing near real time agricultural performance and seasonal outlook analysis.

03

East Africa Drought Watch

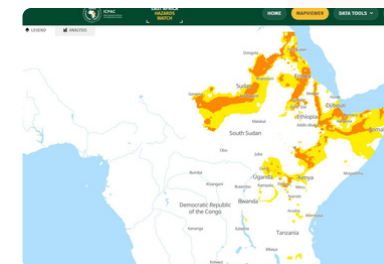
<https://droughtwatch.icpac.net/>



A regional early warning platform for monitoring drought conditions and emerging drought risk across Eastern Africa.

04

East Africa Pest Watch

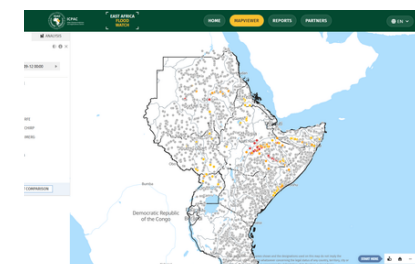


A regional early warning platform for monitoring and warning on potential invasive crop and animal pests.

05

East Africa Flood Watch

<https://floodwatch.icpac.net/>



A regional monitor of extreme flood events such as heavy rainfall, river flooding, and flash floods that can have severe and widespread impacts across the region

Data and information platforms & Digital Innovations

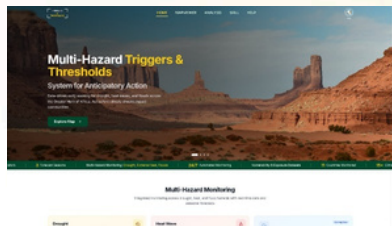


07 **HUSIKA APP**
<https://husika.icpac.net/>



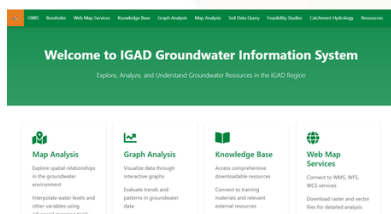
A mobile platform delivering tailored climate and advisory information to farmers and end users.

08 **Multi-Hazards Triggers & Thresholds**
<https://eatriggersthresholds.icpac.net/>



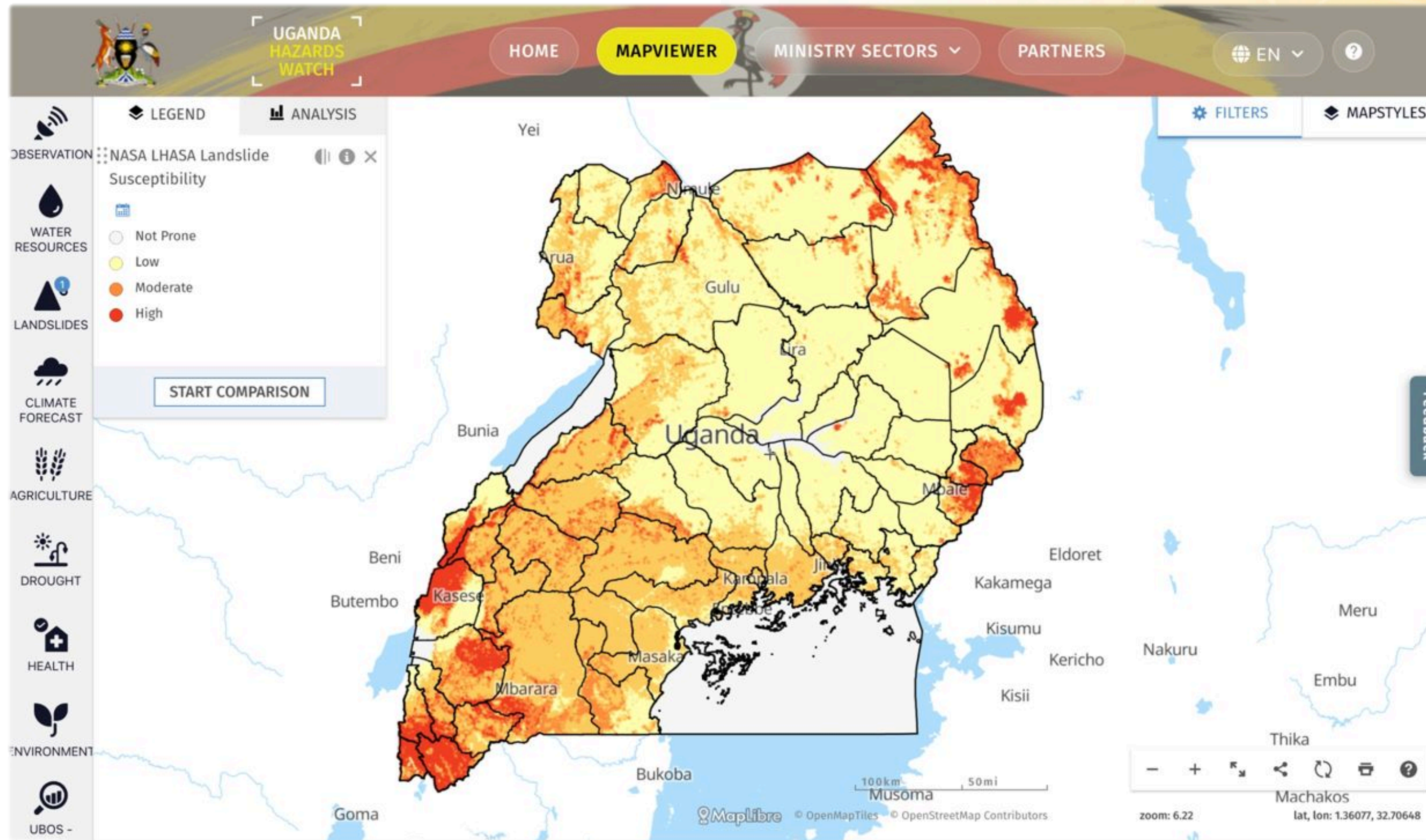
Data-driven early warning for drought, heat waves, and floods across the Greater Horn of Africa. Act before climate shocks impact communities. In support of anticipatory action

09 **IGAD Ground water information System**
<https://water.igad.int/gwis>



Explore, Analyze, and Understand Groundwater Resources in the IGAD Region

► Capacity Building and support



Uganda Multi Hazard Early Warning System

Somali Multi Hazard Watch

SADC Multi Hazard Watch

Data source



The ICPAC services are information driven. Need for different data.

- **Meteorological and hydrological Station data**

- **Model data**

- **Satellite derive estimates data**

RFE2 Satellite rainfall for Africa

ARC2 Long-term average rainfall, used to judge whether recent rain is above or below normal

CHIRPS Satellite rainfall combined with rain gauge readings

GSMaP Satellite rainfall, adjusted using rain gauges

- **Exposure**

LandScan -Where people live

ASAP cropland -Farmland

ASAP rangeland -Grazing land

UNHCR camps -Refugee camps

- **Administrative boundary data**

Administrative boundary

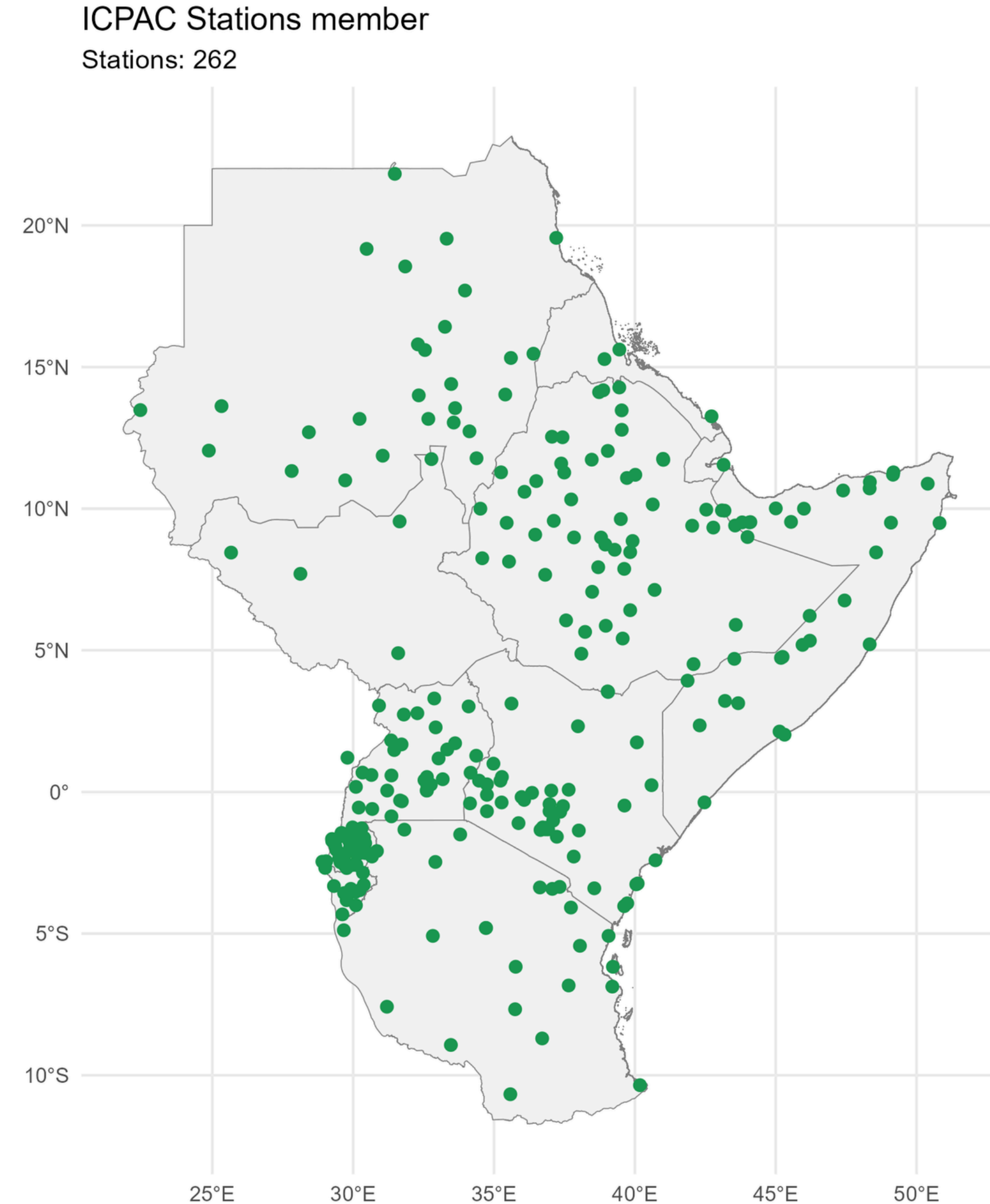
Watershed Boundaries

Protected areas

Rangelands

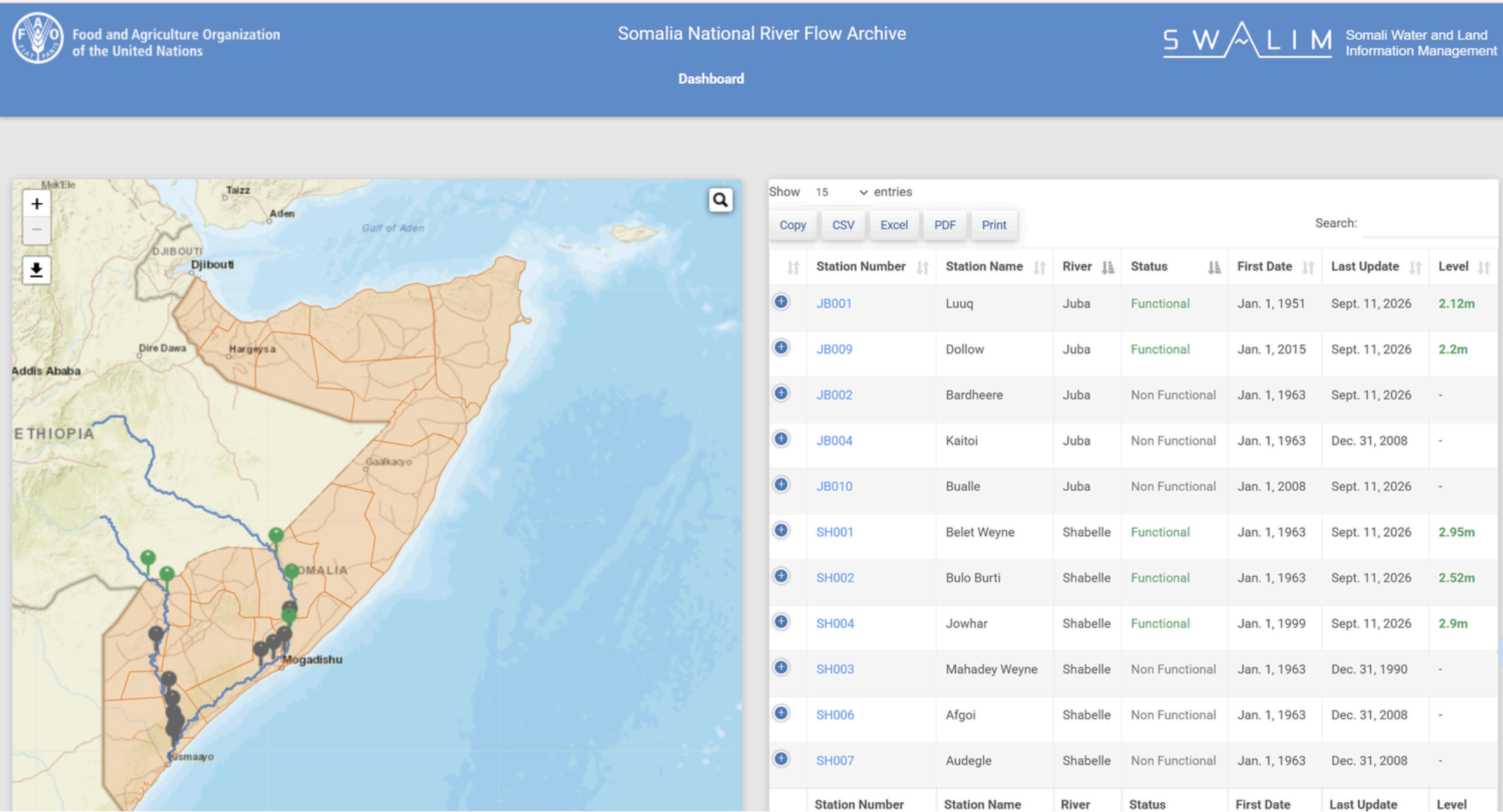
Meteorological Data

- ICPAC receives climate records from 262 stations from the member states.
- Data is shared via the email address data@icpac.net.
- WIS2 to be operationalized- requires members countries to have implemented it.
- Data is processed and stored in the ICPAC CDMS(Climsoft)
- ICPAC has in place a Memorandum of Understanding (MOU) between itself and the member countries on climate data sharing.



Hydrological Data

- ICPAC does not have standing agreement on hydrology data transmission with the member countries.
- Data is either received through a defined purpose or from those that have open access



FAO-SWALIM
Water and land
Information
Management

Rwanda Water
Portal

Challenges in Hydrometeorological data



- General Data Systems Interoperability Issues
- Difference in country's data sharing policies
- Fragmented data systems across countries and institutions, with limited standardization.
- Limited infrastructural, technical and human capacity in some NMHSs.

► Communication, Dissemination & User Engagement



UIPs

Passive Online platforms | Active GHACOF, FSNWG, COP-CPS



Website



Bulletins



ICPAC mailing list



Climate in news briefs



Forums and webinars



Social media



Publications: Reports, papers atlases



ICPAC

Thank you!